Department Overview Department Name: Department of Geological Engineering Faculty: Faculty of Engineering & Architecture Accreditation: Officially accredited by the Pakistan Engineering Council (PEC). History: Initiated in 2008 as the first degree-awarding Geological Engineering program under the engineering faculty at BUITEMS. It was established to address the critical need for hidden natural resource exploration and economic development in Balochistan, which is Pakistan's largest and most mineral/hydrocarbon-rich province (with roughly 90% of its area historically unexplored). Core Technical Domains: Analyzing the engineering behavior of rock, soil, and geotechnics against constructed projects, and guiding mineral and hydrocarbon discovery. Career Sectors: High demand in the energy and construction sectors. Graduates find roles as Geotechnical Engineers, Geologists, Construction Consultants, or Civil Contractors across public and private organizations. Key employment fields include petroleum/gas organizations, atomic energy, the Irrigation Department, academic institutions, and Natural Energy Resources Exploration.

Mission The Department of Geological Engineering is committed to train engineers with advanced knowledge and contemporary skills, while promoting a culture of research, equipping students with the art of living to achieve success in their future professional careers not only in Pakistan but throughout the world.

Program Educational Objectives (PEOs) Graduates are expected to demonstrate the following attributes 3 to 5 years post-graduation:

1. The ability to apply technological and engineering fundamentals for the solution of geotechnical and environmental problems that are related to Geological processes.
2. The ability to exhibit effective communication, teamwork, leadership skills, and technical competence enabling them to excel in their professional careers as well as in higher studies.
3. The ability to fulfill ethical, professional, and social responsibilities in their professional lives and community services.

Bachelor of Science in Geological Engineering (BS Geological Engineering)

Admission Requirements • Academic Background: F.Sc (Pre-Engineering / Pre-Medical / ICS) from any recognized board with at least 60% marks, OR a Diploma of Associate Engineer (DAE) in Civil, Land and Mine Surveying, or Mining securing a minimum of 60% marks. • Testing: Candidates must successfully qualify for the university admission test.

Degree Requirements Total Credit Hours: 136 Total Courses: 46 Minimum Academic Standing: CGPA $\ge$ 2.0 PLO Requirement: The program adopts the 12 expected standard engineering graduate attributes defined in the PEC Accreditation Manual 2014 (Engineering Knowledge, Problem Analysis, Design/Solutions, Investigation, Modern Tool Usage, Engineer & Society, Sustainability, Ethics, Teamwork, Communication, Project Management, and Lifelong Learning).

Curriculum Plan (Program Schema by Semester) Note: Credit hour structures denote (Theory + Lab).

1st Semester • ENGG-201/ENGG-111 | Engineering Drawing & Graphics / Basic Electrical & Electronics | Credit Hours: 1+1 / 2+1 • GEOE-203/CHE-102 | Physical Geology / Applied Chemistry | Credit Hours: 3+1 / 2+0 • MATHP-118/HUM-101 | Calculus & Analytical Geometry / Islamic Studies or Ethics | Credit Hours: 3+0 / 2+0 • HUM-163 | Functional English | Credit Hours: 2+0

2nd Semester • GEOE-305/GEOE-202 | Mineralogy & Petrology / Intro to Geological Engineering | Credit Hours: 2+1 / 2+0 • MATHP-214/PHY-205 | Differential Equations / Applied Physics | Credit Hours: 3+0 / 2+0 • Hum-268/HUM-102 | Communication Skills / Pak-Studies | Credit Hours: 2+0 / 2+0 • IT-102/ENIRON-305 | Intro to ICT / Occupational Health & Safety | Credit Hours: 2+* / 1+0 *(Note: Lab component incomplete in raw text layout list)

3rd Semester • ENGG-215/MINE-201 | Applied Thermodynamics / Stratigraphy & Structural Geology | Credit Hours: 2+1 / 2+1 • MATHP-111/ENGG-206 | Linear Algebra / Fluid Mechanics | Credit Hours: 3+0 / 3+1 • CS-112/ENGG-241 | Computer Programming / Engineering Economics | Credit Hours: 2+1 / 2+0

4th Semester • GEOE-206/GEOE-208 | Mechanics of Material / Surveying | Credit Hours: 3+1 / 2+2 • CS-115/GEOE-312 | Computer Science & Numerical Analysis / Hydrogeology | Credit Hours: 2+0 / 3+1 • STAT-101/HUM-276 | Probability & Statistics / Tech Report Writing & Pres. Skills | Credit Hours: 2+0 / 2+0

5th Semester • GEOE-302/GEOE-401 | Engineering Geology / Intro to Geophysical Exploration Tech | Credit Hours: 2+1 / 3+1 • GEOE-311/MINE-307 | Geotechnical Engineering-I / Explosives Engineering | Credit Hours: 3+1 / 2+1 • MGMT-211/SOC-301 | Engineering Management / Sociology for Engineers | Credit Hours: 2+0 / 2+0

6th Semester • GEOE-313/MINE-401 | Petroleum Geology / Rock Mechanics | Credit Hours: 2+1 / 3+1 • GEOE-403/GEOE-316 | Earth Quake Seismology & Risk Assessment / GIS & Remote Sensing | Credit Hours: 2+1 / 2+1 • GEOE-417 | Environmental Geological Engineering | Credit Hours: 2+1

7th Semester • TBA / MINE-404 | Major Elective I / Tunneling & Excavation Engineering | Credit Hours: 3+1 / 3+1 • TBA / TBA | Major Elective II / Drilling Engineering | Credit Hours: 2+1 / 2+1 • GEOE-410L | Project - I | Credit Hours: 0+2

8th Semester • TBA / TBA | Major Elective III / Major Elective IV | Credit Hours: 3+1 / 2+1 • MGMT-201/GEOE-410L | Entrepreneurship / Project - II | Credit Hours: 3+0 / 0+4 • Grand Total: 136 Credit Hours

Major Electives Curriculum (5th Board of Studies) The program offers 24 elective courses partitioned into three distinct professional specializations under the current curriculum framework. Students choose 4 courses in total from one preferred stream (2 courses in the 7th semester, and 2 courses in the 8th semester): • GRE: Geotechnical and Rock Engineering • NERE: Natural Energy Resource Exploration • GEE: Geo-Environmental Engineering Specialization Streams Course Options

Major Elective-I (Semester 7 - 3+1 Credits): • GRE: Geotechnical Engineering-II OR Statistical Methods in Geology and Engineering • NERE: Seismic Data Processing & Interpretation OR Ground water Resource Engineering • GEE: Environment Geology & Waste Management OR Environmental Aspects of Mining

Major Elective-II (Semester 7 - 2+1 Credits): • GRE: Geotechnical Site Investigation OR Fluvial Processes • NERE: Reservoir Engineering OR Surficial Processes, Sedimentation and Stratigraphy • GEE: Risk Assessment in Environmental Studies OR Exploration and Environmental Geochemistry

Major Elective-III (Semester 8 - 3+1 Credits): • GRE: Pavement and Foundation Engineering OR Geomorphology and Terrain Analysis • NERE: Petrophysics and Well Logging OR Petrogenesis

and Metallogenesis • GEE: Environmental Soil Science OR Site Remediation Engineering
Major Elective-IV (Semester 8 - 2+1 Credits): • GRE: Ground Improvement and Geosynthetics OR Analysis of Rock Structures • NERE: Production Engineering OR Carbonate Sedimentology • GEE: Environmental Data Analysis OR Recycling and Sustainable Engineering